



POLITÉCNICA

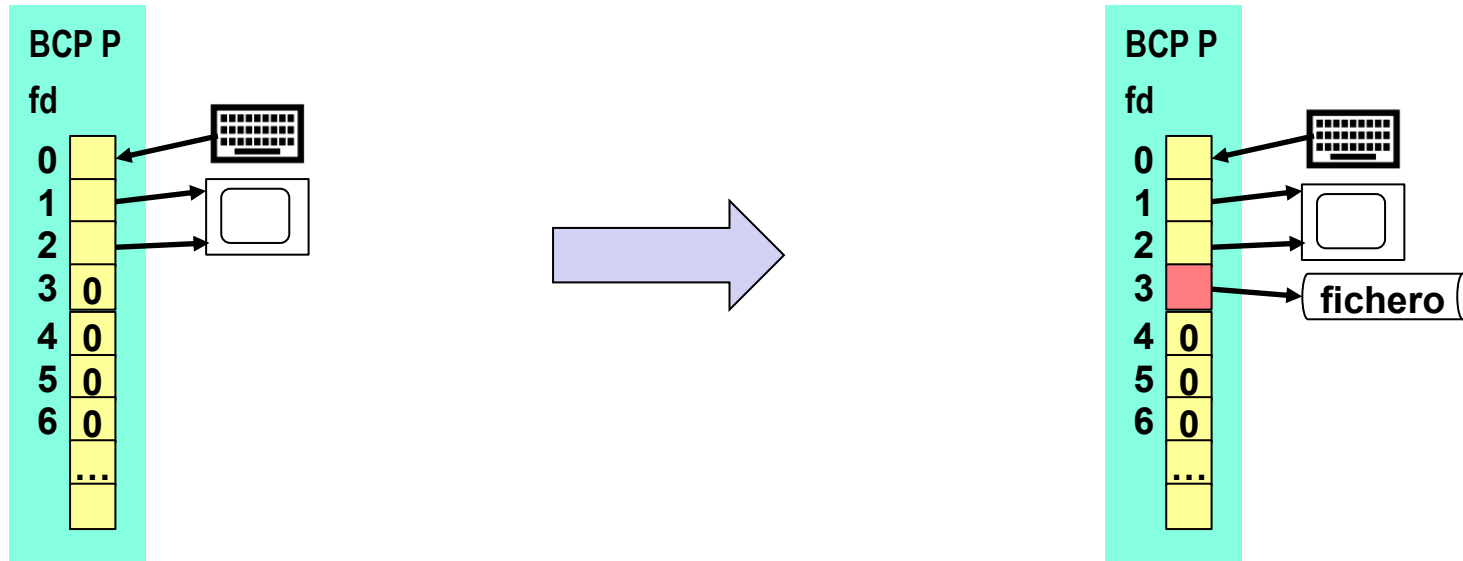
REDIRECCIONES

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Curso 2026/27

Programa que ejecuta: ls > fichero

```
int main(void) {
    pid_t pid;
    int status, fd;
    fd = open("fichero", O_WRONLY|O_CREAT|O_TRUNC, 0666);
    if (fd < 0) {perror("open"); exit(1);}
    pid = fork();
    switch(pid) {
        case -1:          /* error */
            perror("fork"); exit(1);
        case 0:           /* proceso hijo ejecuta "ls" */
            close(1);
            dup(fd);      /* el descriptor se copia en el 1 */
            close(fd);
            execlp("ls", "ls", NULL);
            perror("execlp"); exit(1);
        default:          /* proceso padre */
            close(fd);
            while (pid != wait(&status));
    }
    return 0;
}
```

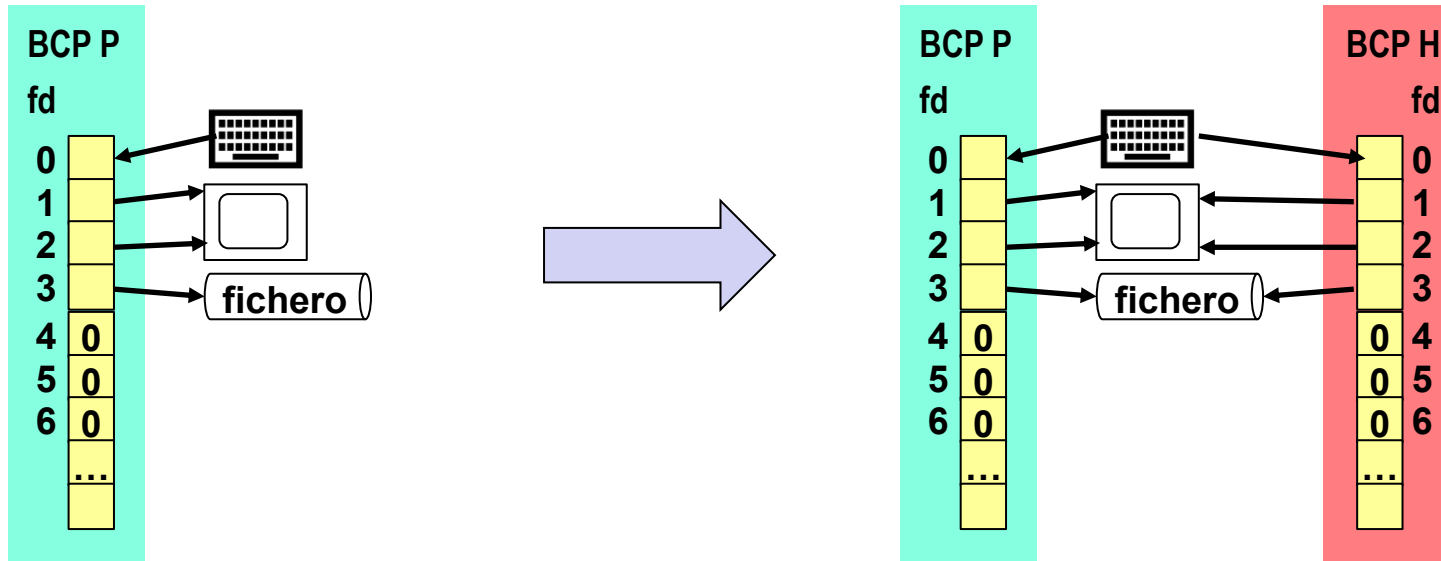
ls > fichero CON DUP (PASO 1)



Proceso P: $fd = \text{open}(\text{"fichero"}, \dots) \rightarrow fd = 3$

ls > fichero CON DUP (PASO 2)

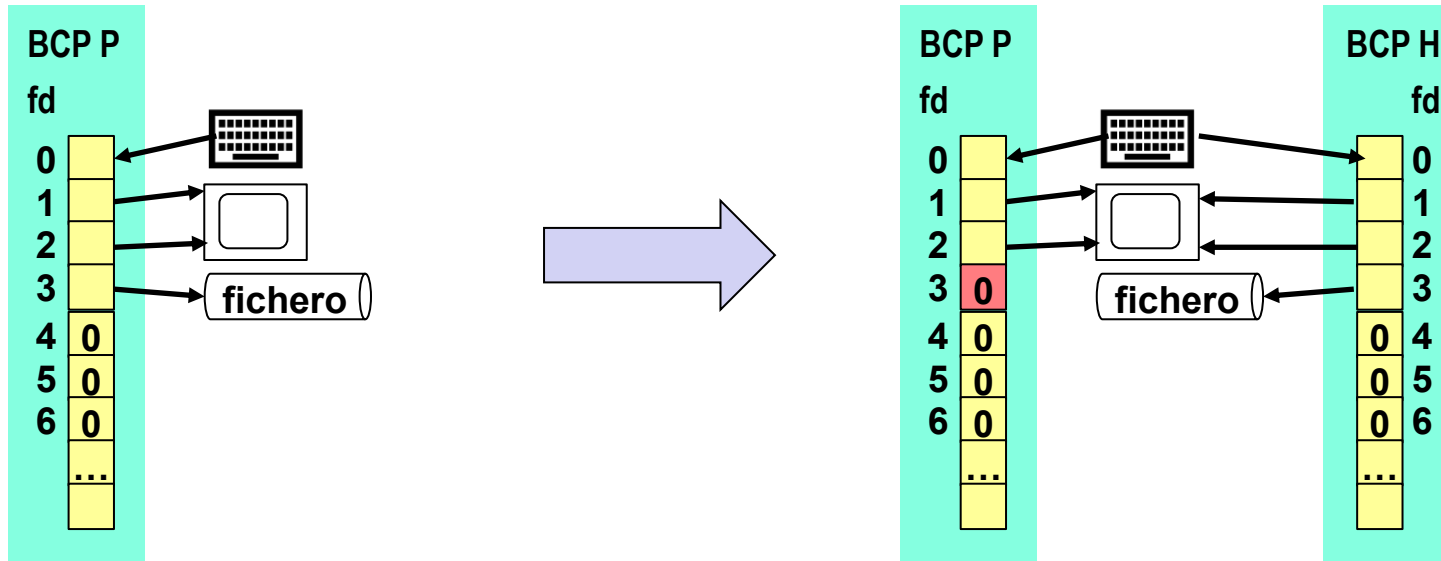
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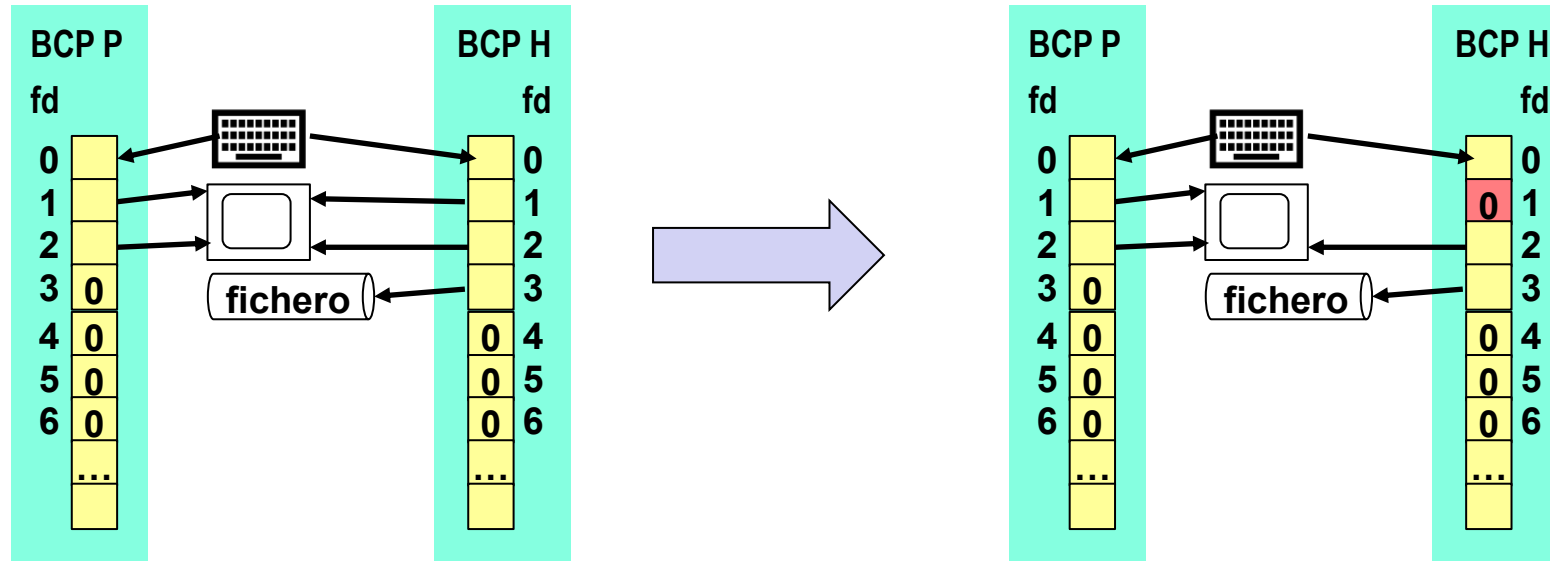
Proceso P: *fork()* → Proceso H

Is > fichero CON DUP (PASO 3)

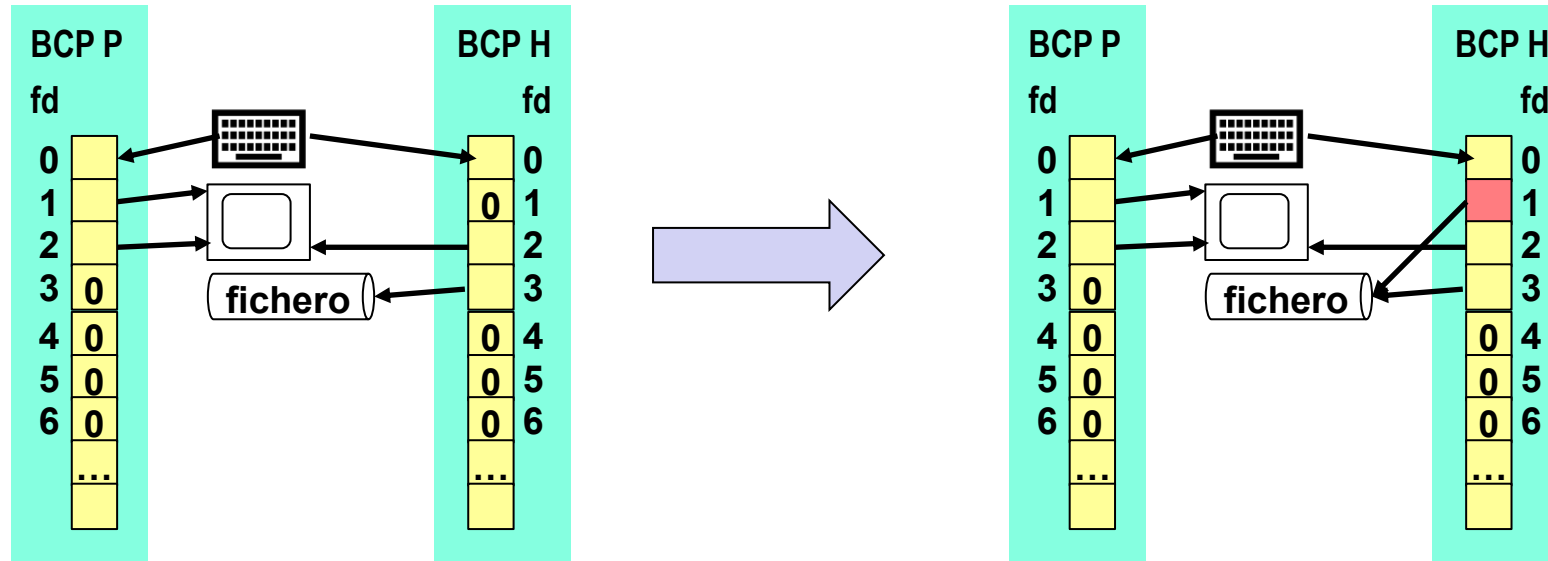
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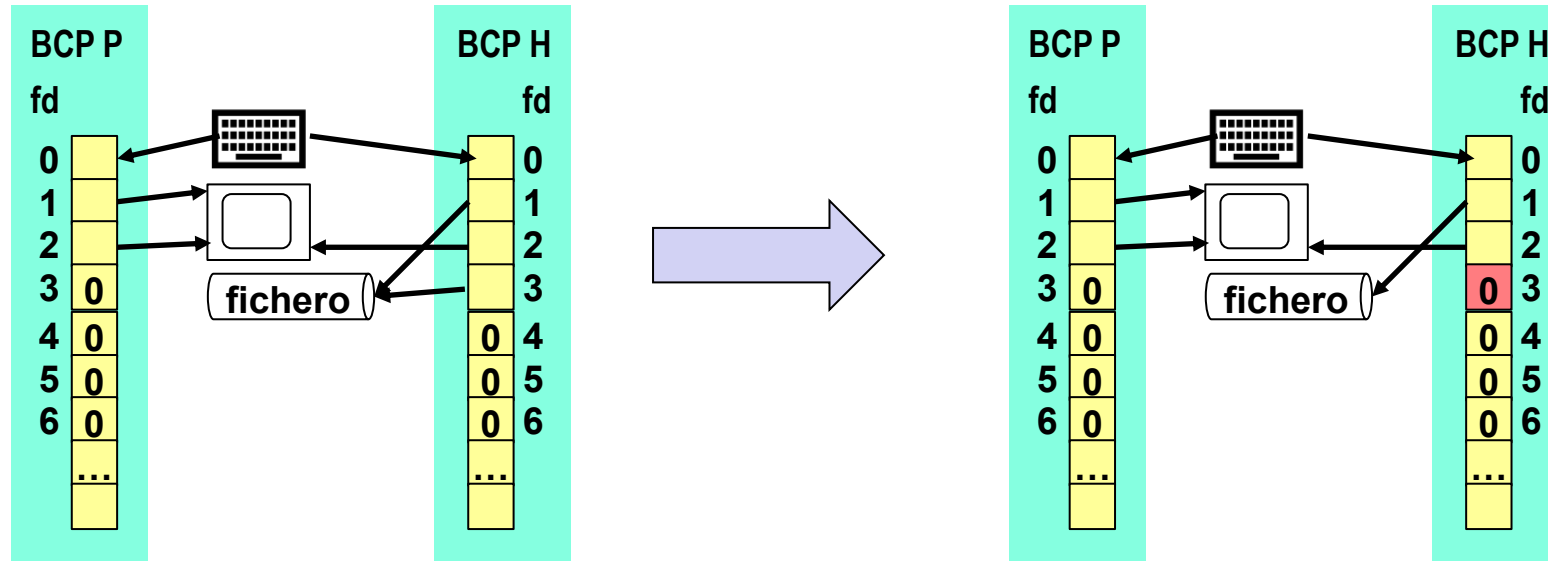
Proceso P: `close(fd)` // (fd = 3)



Proceso H: `close(1)`



Proceso H: *dup(fd) // (fd = 3)*



Proceso H: `close(fd)` // (fd = 3)

Programa que ejecuta: ls > fichero

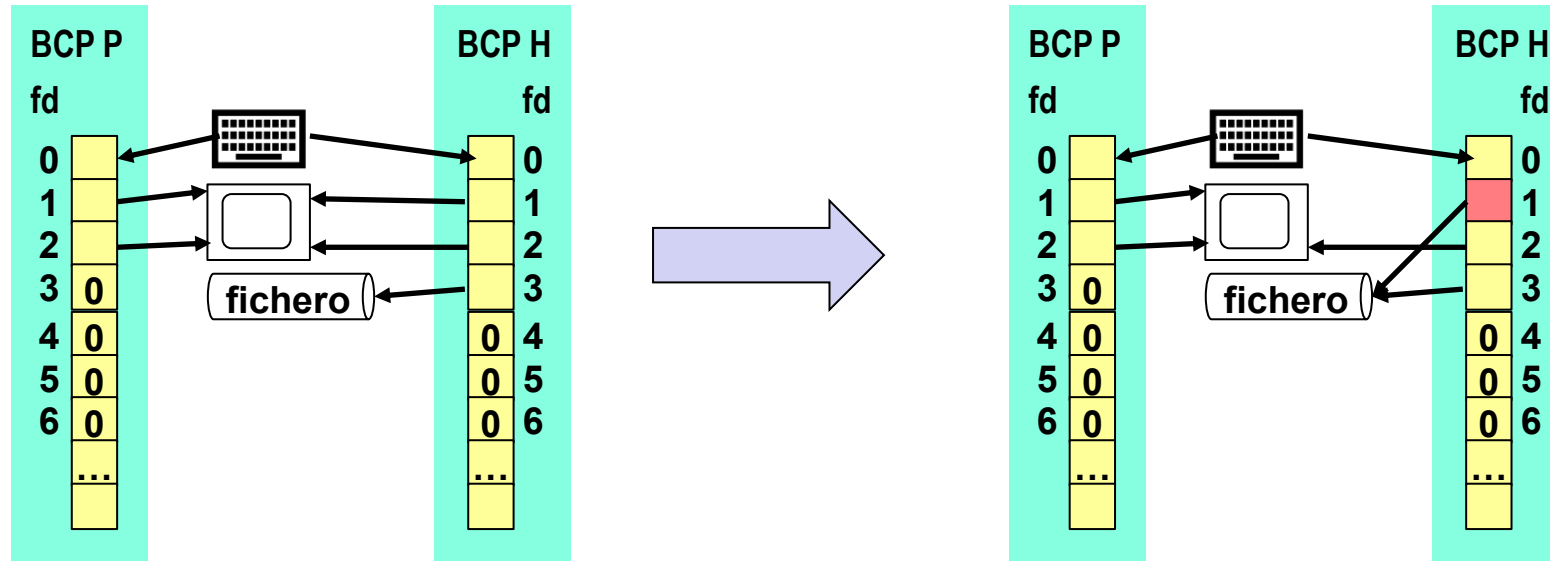
```
int main(void) {
    pid_t pid;
    int status, fd;
    fd = open("fichero", O_WRONLY|O_CREAT|O_TRUNC, 0666);
    if (fd < 0) {
        perror("open"); exit(1);
    }
    pid = fork();
    switch(pid) {
        case -1:                /* error */
            perror("fork"); exit(1);
        case 0:                /* proceso hijo ejecuta "ls" */
            dup2(fd, 1); /* el descriptor se copia en el 1 */
            close(fd);
            execlp("ls", "ls", NULL);
            perror("execlp");
            exit(1);
        default:                /* proceso padre */
            close(fd);
            while (pid != wait(&status));
    }
    return 0;
}
```

Is > fichero CON DUP2 (PASO 4; FUSIÓN PASOS 4 Y 5 PREVIOS)

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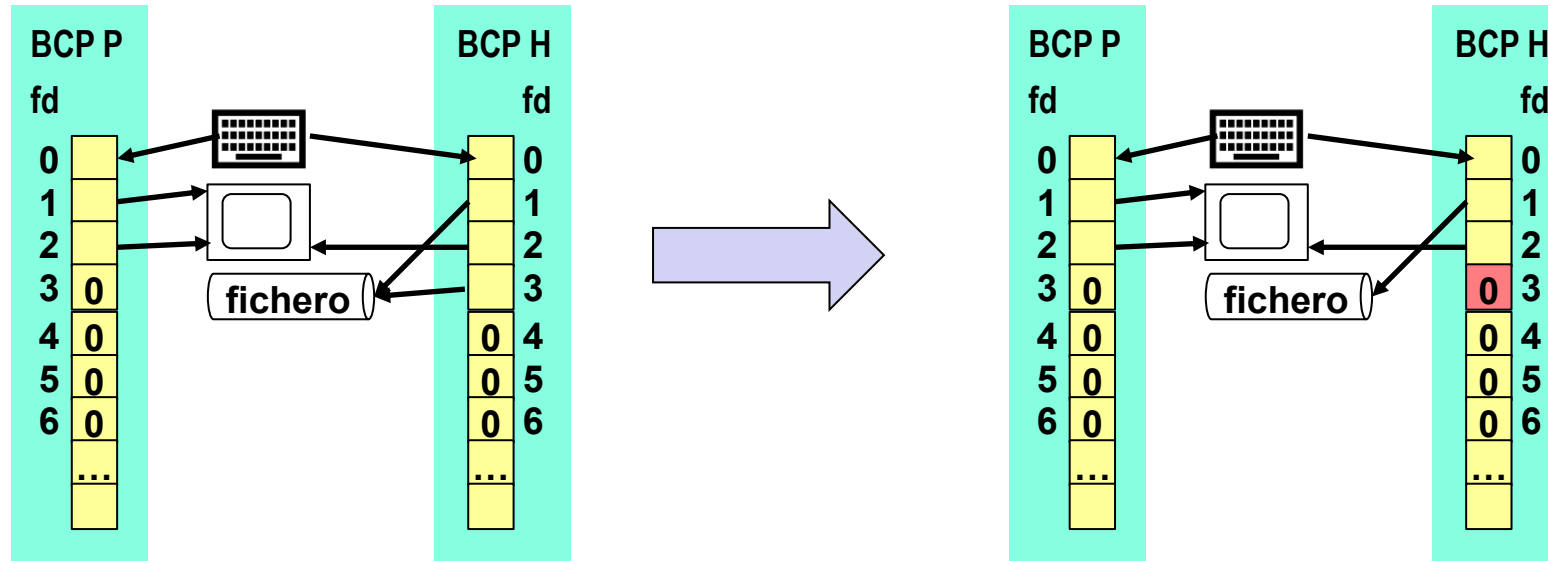
POLITÉCNICA



Proceso H: `dup2(fd, 1) // (fd = 3)`

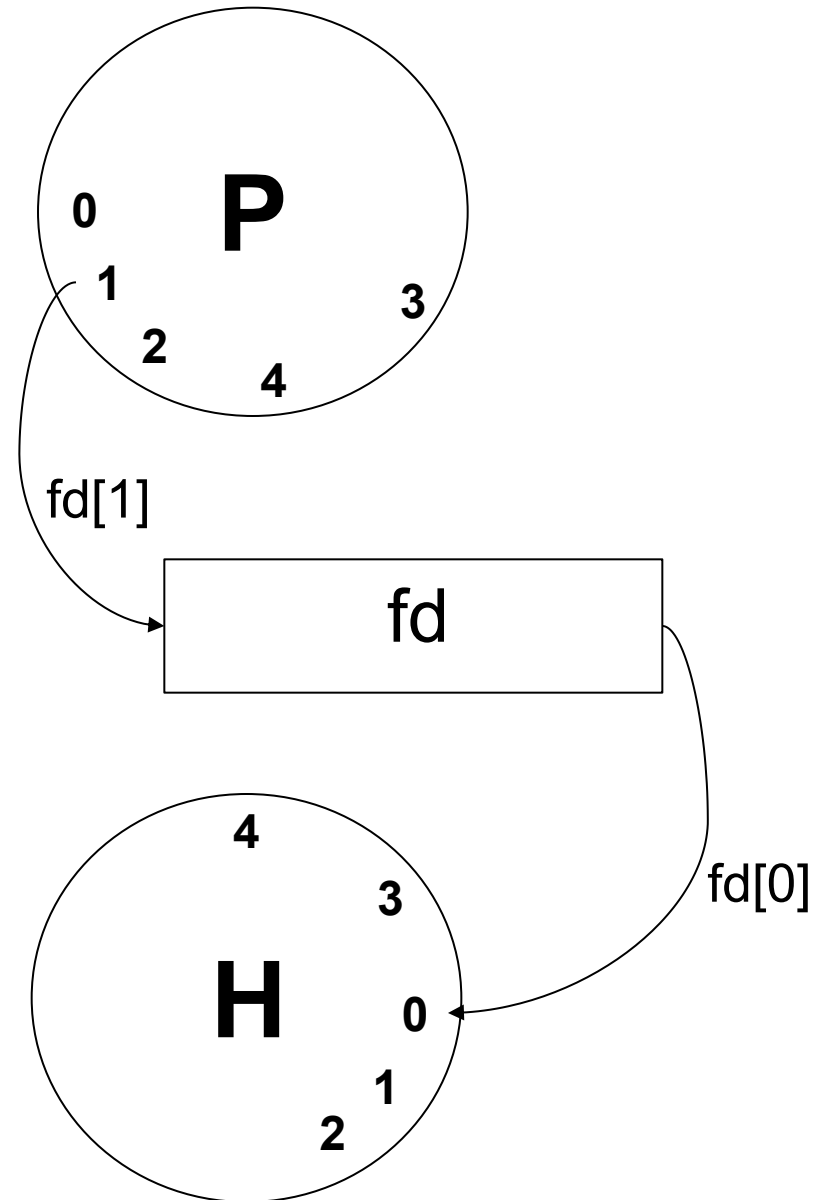
Is > fichero CON DUP2 (PASO 5; IGUAL QUE PASO 6 PREVIO)

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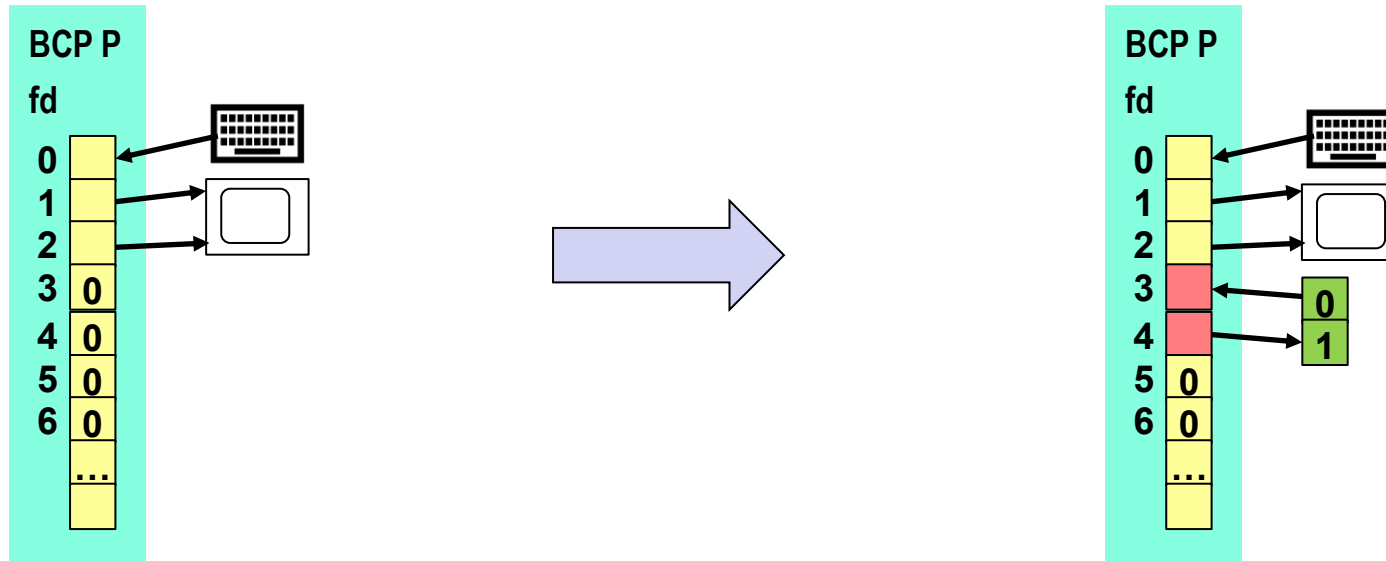
Proceso H: `close(fd)` // (fd = 3)

```
int fd[2], ret;
ret = pipe(fd);
pid = fork();
if (pid > 0) // padre
{
    close(1);
    dup(fd[1]);
    close(fd[0]);
    close(fd[1]);
}
else // hijo
{
    close(0);
    dup(fd[0]);
    close(fd[0]);
    close(fd[1]);
}
```

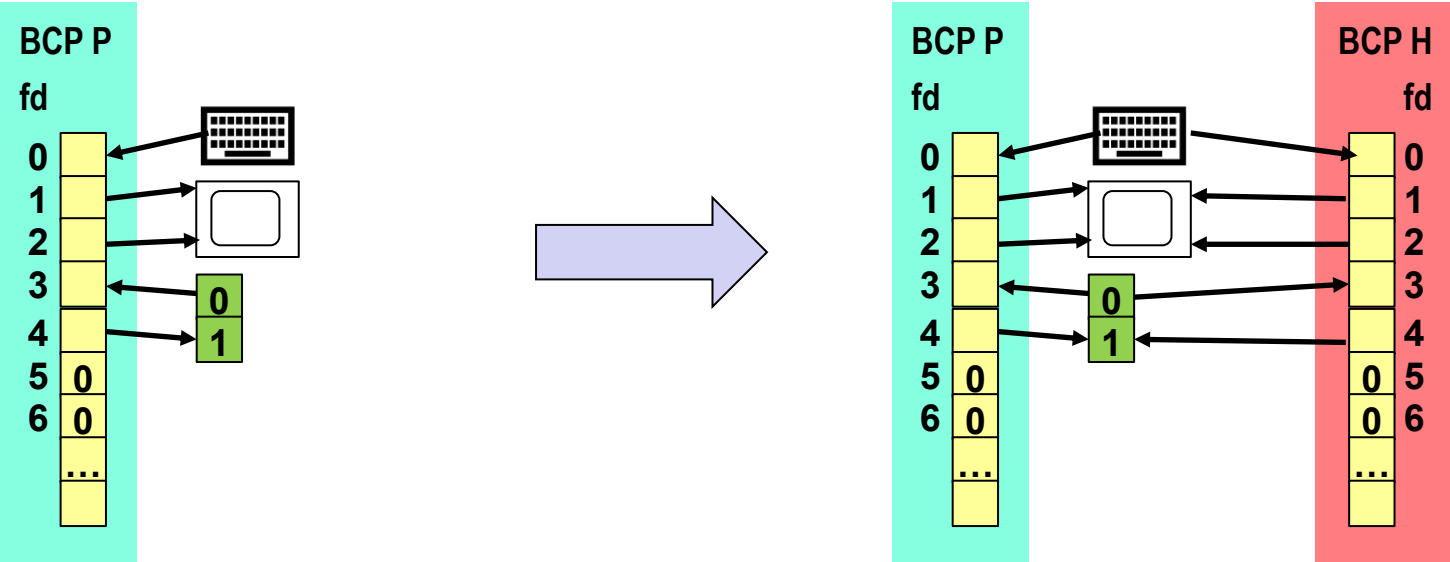


REDIRECCIÓN CON PIPE (PASO 1)

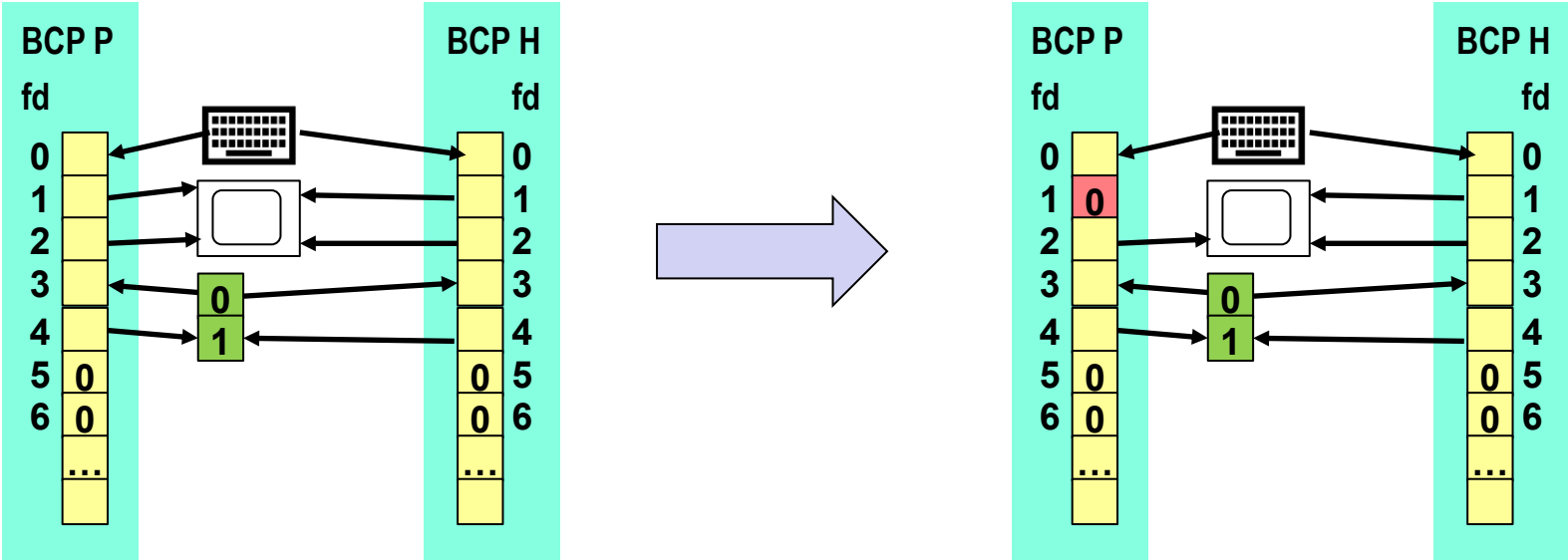
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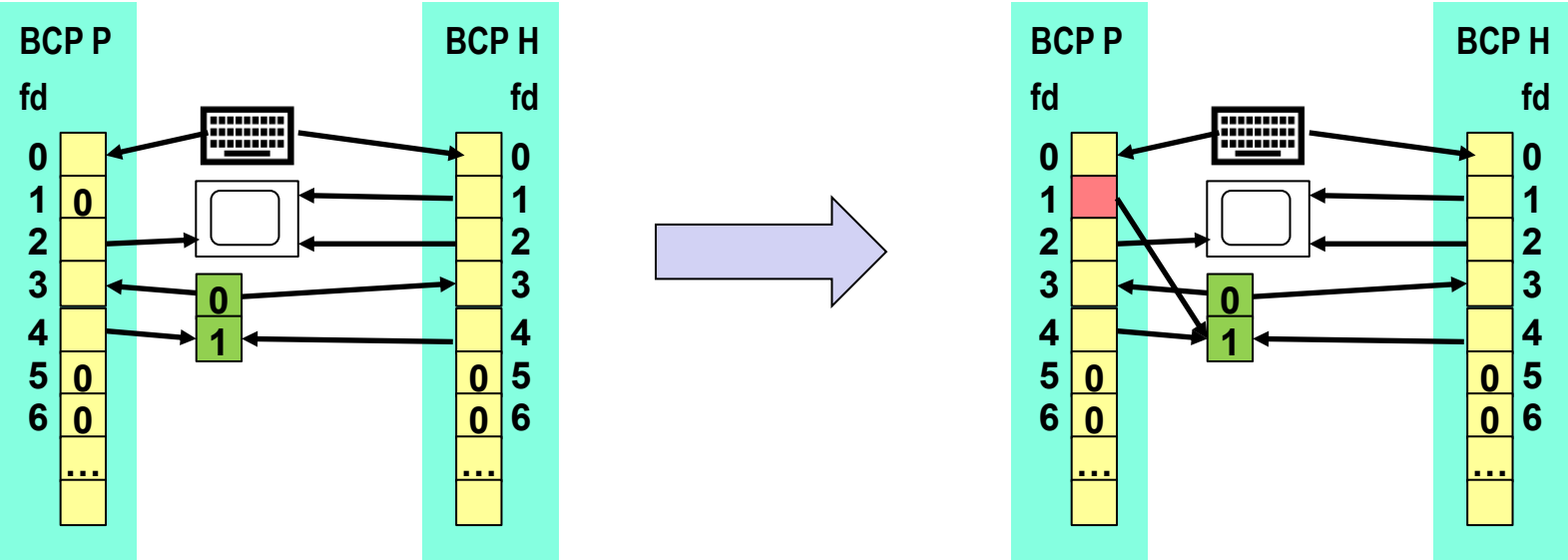
Proceso P: $\text{pipe}(fd) \rightarrow fd = [3,4]$



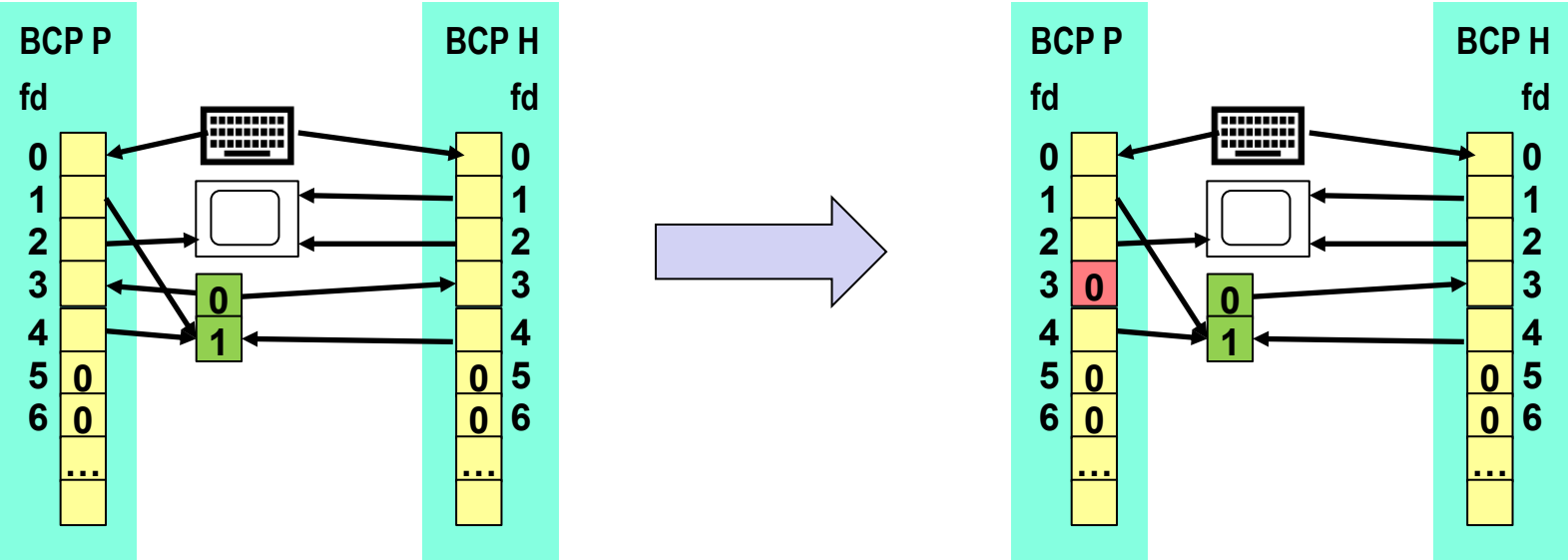
Proceso P: *fork()* → Proceso H



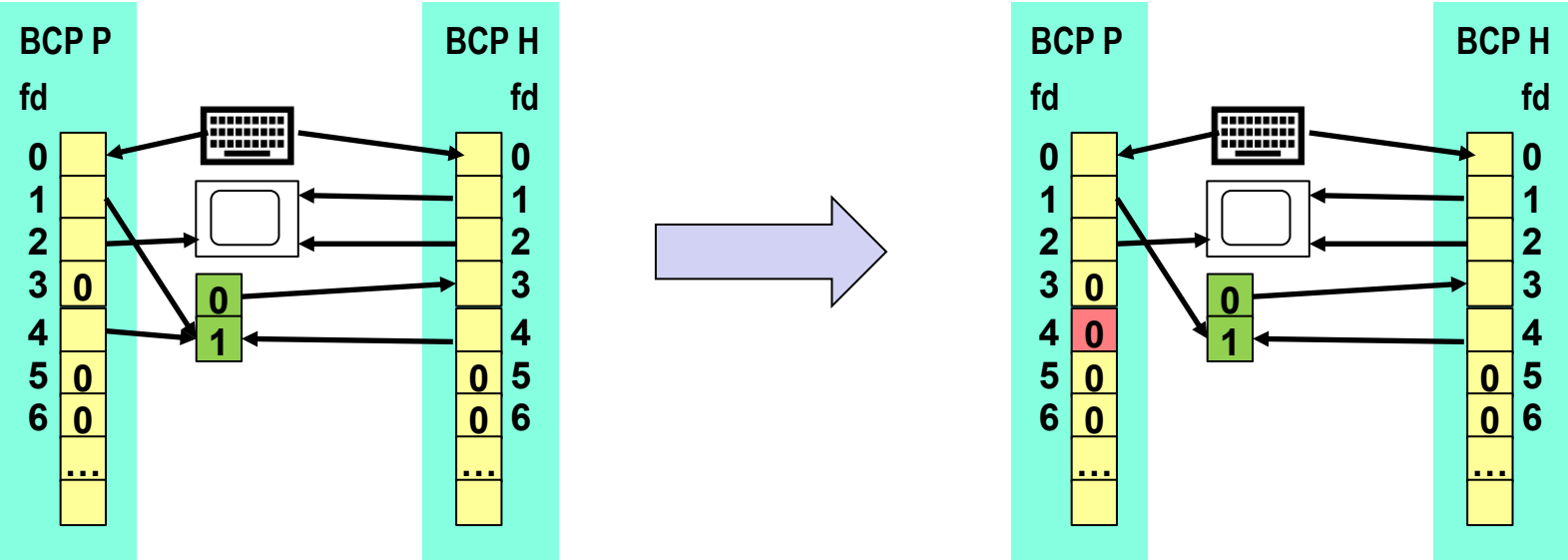
Proceso P: *close(1)*



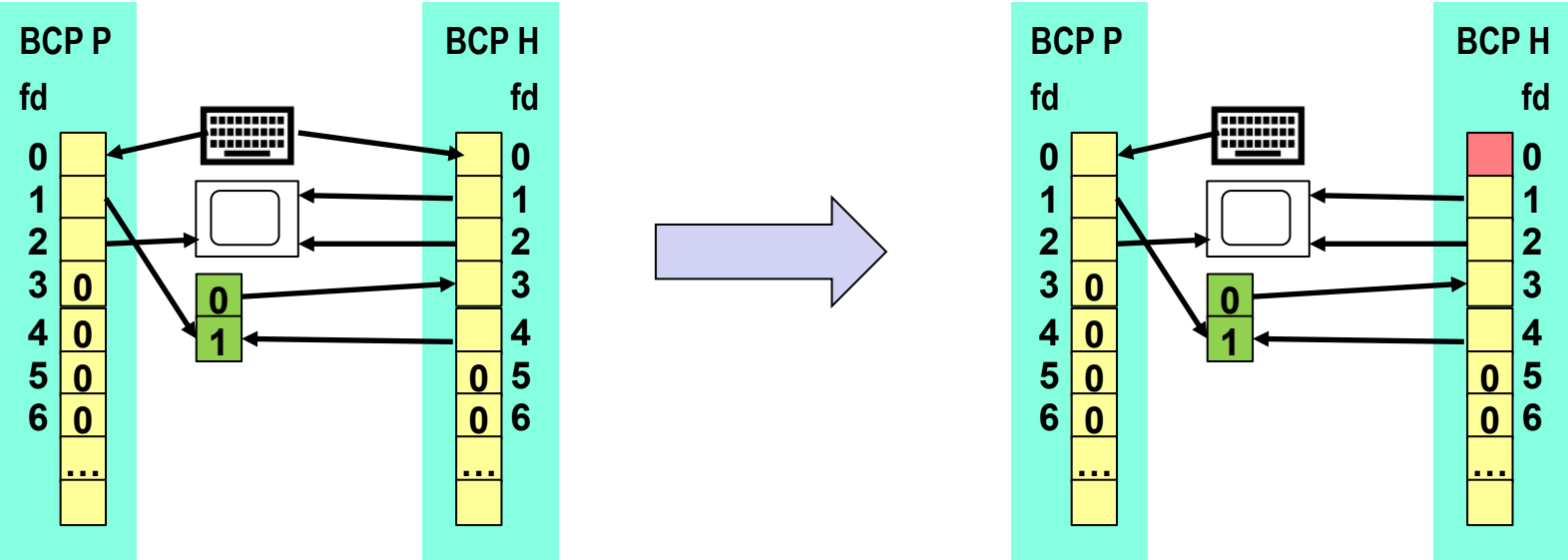
Proceso P: `dup(fd[1])`



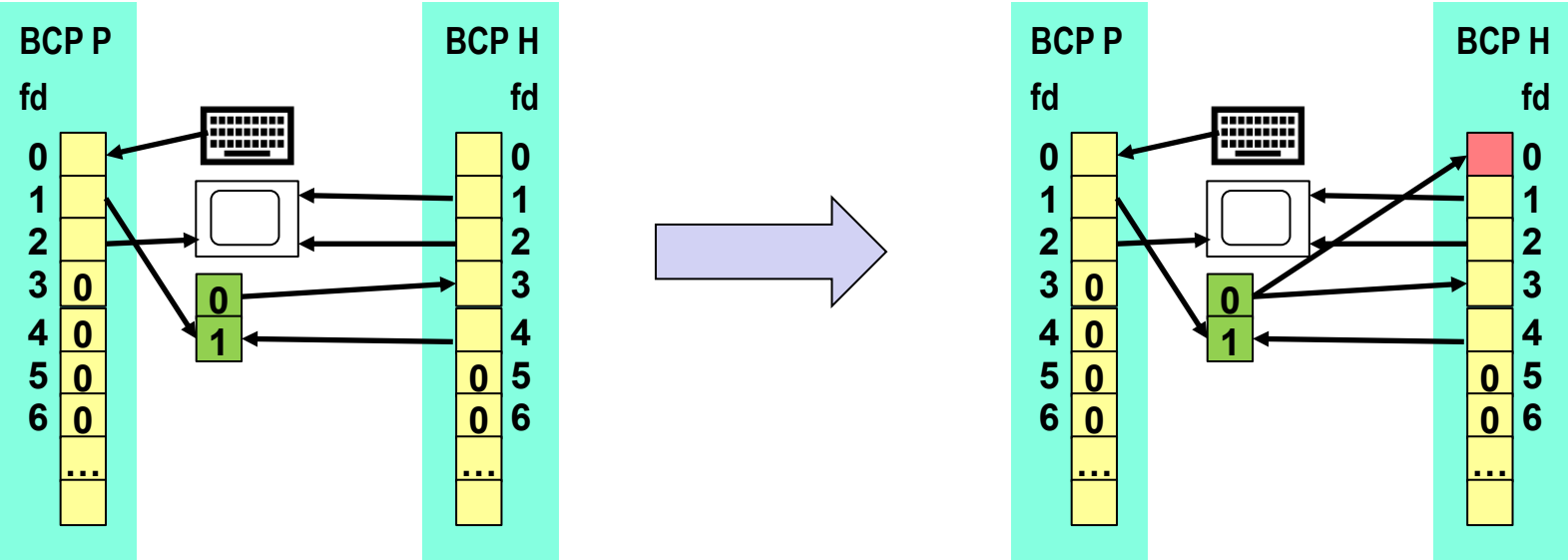
Proceso P: `close(fd[0])`



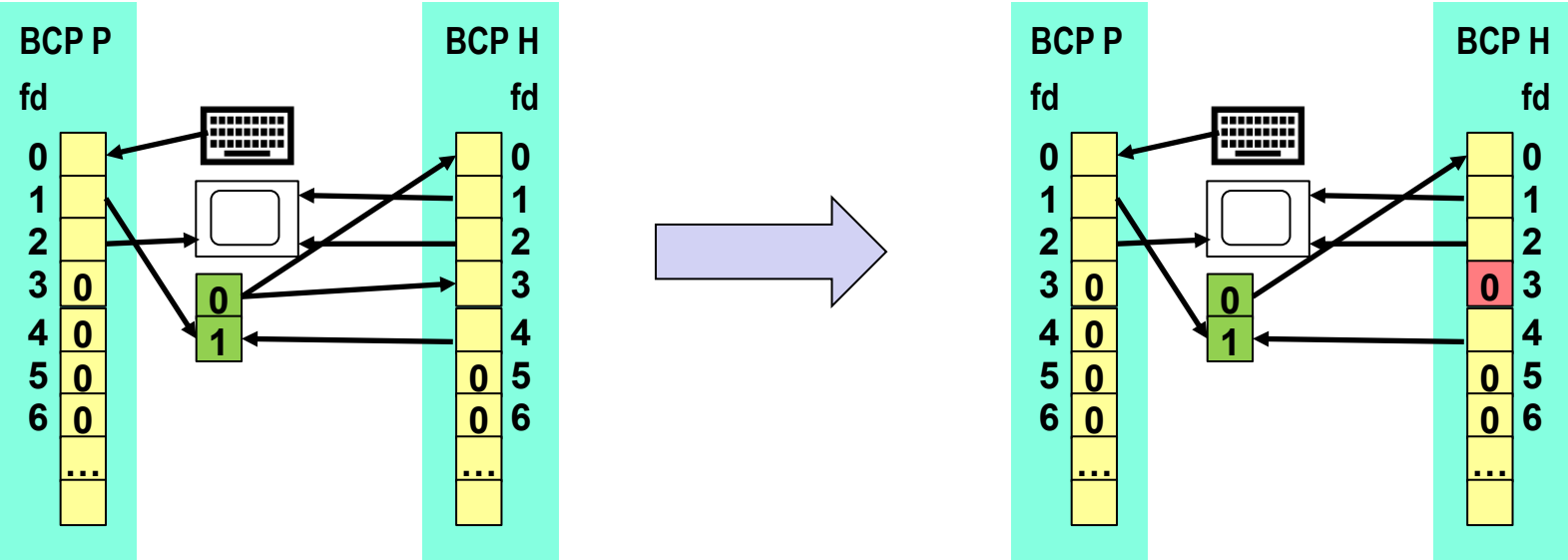
Proceso P: `close(fd[1])`



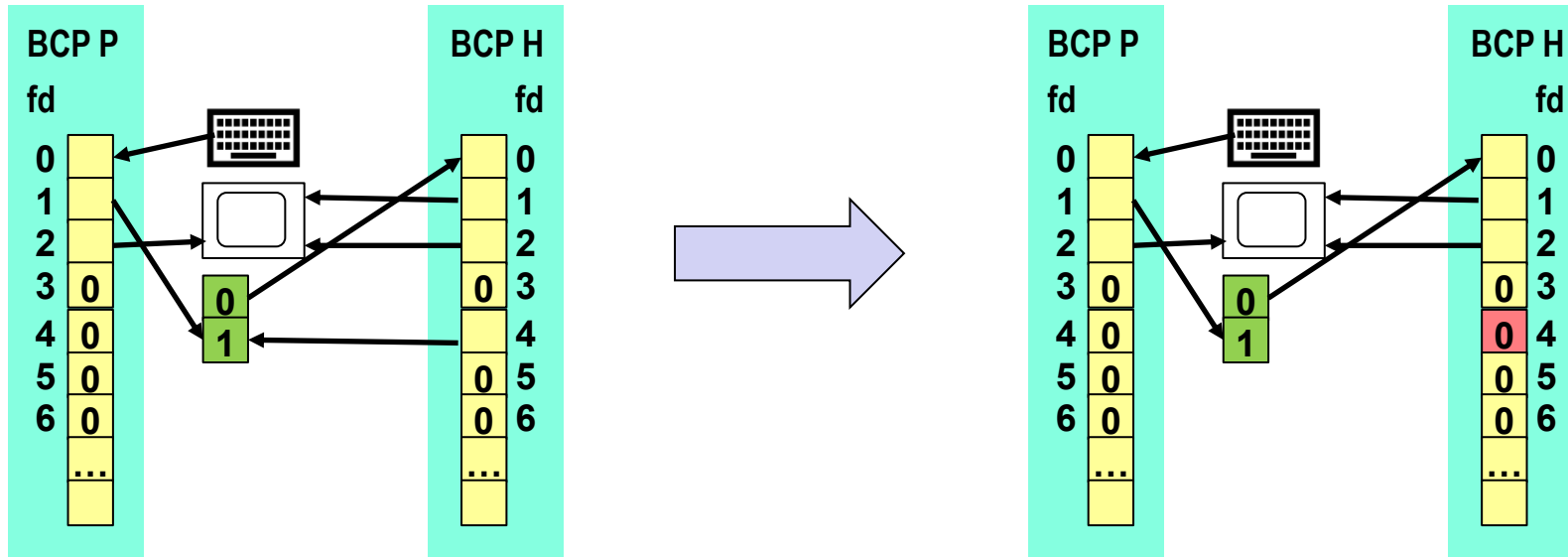
Proceso H: `close(0)`



Proceso H: `dup(fd[0])`



Proceso H: `close(fd[0])`



Proceso H: `close(fd[1])`